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SOLVEWISE: AN ADAPTIVE WEB PLATFORM FOR PERSONALIZED MATHEMATICS REMEDIATION IN MIDDLE SCHOOL

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Abstract:

Educational outcomes in Bulgaria continue to be a significant concern. According to the 2022 PISA results, Bulgarian 15-year-olds scored an average of 417 points in mathematics, with approximately 53.6% of students failing to reach the minimum proficiency level. To address this challenge, the present report introduces SolveWise, a digital educational platform designed to support 7th-grade students in overcoming learning gaps in mathematics through an adaptive remediation approach.

The project demonstrates the potential of adaptive learning in a school context by combining lightweight web technologies with data-driven instructional design. Expected benefits include increased student engagement, gradual development of confidence in mathematical skills, and improved overall performance. Planned future enhancements include the deployment of self-learning algorithms and the expansion of mistake-pattern categories, error-pattern analysis, gamification, AI, and potential applicability in additional school subjects.

1. INTRODUCTION

The middle school years represent a critical period for the development of mathematical literacy, yet many students encounter significant challenges with concepts such as fractions, linear equations, and geometric principles (National Mathematics Advisory Panel, 2008). Traditional, one-size-fits-all instructional methods often fail to address the diverse learning paces and needs within a single classroom, leading to knowledge gaps that can hinder future academic achievement.

Digital learning environments have the potential to significantly support students who struggle with mathematics, especially through individualized learning pathways. Yet, most of the currently available platforms face two major limitations: many are financially inaccessible for large-scale implementation in schools, and others focus mainly on repetitive exercises without addressing deeper conceptual understanding. What is increasingly needed are systems that go further, such as tools that combine assessment with hands-on exploration of mathematical ideas, while

also giving both students and teachers clear insight into learning progress and areas that require reinforcement.

This paper introduces SolveWise, a web platform designed to bridge this gap. SolveWise is founded on the pedagogical principles of constructivism, where learning is facilitated through interactive exploration (Piaget, 1973), and mastery learning, which emphasizes the importance of demonstrating proficiency before advancing (Bloom, 1968). The platform's primary objectives are:

- To deploy a suite of interactive lessons that use dynamic manipulation to illustrate core mathematical concepts.
- To provide an adaptive testing environment that personalizes practice based on student performance.
- To implement an automated, visual progress-tracking system that empowers students to take ownership of their learning journey.

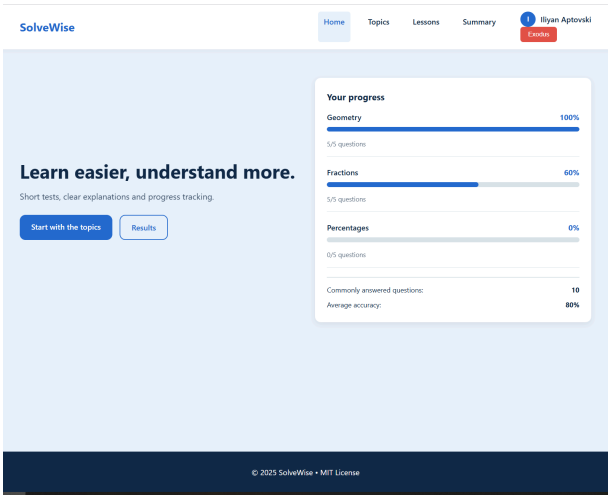


Fig. 1 Home Page

2. RELATED WORK

The landscape of educational technology for mathematics is extensive, ranging from basic flashcard apps to advanced Intelligent Tutoring Systems (ITS). Platforms such as **Carnegie Learning's MATHia** and **ALEKS** employ sophisticated cognitive models and knowledge space theory to generate highly personalized learning paths (Falmagne et al., 2006). Although effective, their complexity and subscription-based cost structures often create barriers for widespread adoption, especially in regions with limited budgets. Additionally, as far as current offerings indicate, these systems are not available in Bulgarian, which limits accessibility for many students and educators.

Open-source and freely accessible platforms like **Khan Academy** have shown the global impact that video-based instruction paired with practice exercises can achieve. However, their primary focus remains on English-speaking audiences, and the interactive elements are largely limited to multiple-choice tasks or numeric input, providing fewer opportunities for deep conceptual manipulation that modern web technologies can support.

SolveWise positions itself deliberately within this space by serving a specific and currently underserved niche: lightweight, highly interactive visualization tools for core middle school mathematics. Unlike complex ITS platforms, **SolveWise** emphasizes transparent progress tracking

3. METHODOLOGY AND SYSTEM ARCHITECTURE

SolveWise was developed using an Agile methodology, with iterative cycles of design, implementation, and testing. The core architectural pattern is a client-server model, where the client is a static web application, and the server is provided by **Google's Firebase platform**.

3.1 CONCEPTUAL SYSTEM ARCHITECTURE

The system architecture can be conceptualized in three primary layers:

Presentation Layer (Client-Side): This layer comprises the HTML, CSS, and JavaScript files that render the user interface in the student's web browser. It is responsible for displaying interactive lessons, quiz interfaces, and progress dashboards. All user interactions (clicking, dragging, and inputting answers) are handled here.

Application Logic Layer (Client & Server-Side): Client-Side Logic: JavaScript modules manage the application state, including the current lesson, quiz progress, and user inputs. They also handle the fetching of static content from the web server (e.g., lesson JSON files).

Server-Side Logic (via Firebase): Firebase Authentication manages user sessions and role-based access. Firebase Realtime Database rules enforce data security, ensuring users can only read and write their own data.

Data Layer: This layer is divided into two parts:

- **Static Content Data:** JSON files hosted with the web application contain the structure and content for all topics, lessons, and quiz questions. This separation allows for easy content updates without code changes.
- **Dynamic User Data:** The Firebase Realtime Database stores all student-generated data, including quiz scores, mistake histories, and progress metrics.

3.2 STUDENT INTERACTION FLOWCHART

The user journey through the platform is designed for intuitive navigation:

- **Authentication Gateway:** The user arrives at the landing page and must register or log in. The system authenticates them via Firebase Auth and redirects them to the main dashboard.
- **Dashboard & Topic Selection:** The student views their overall progress and selects a mathematical topic (e.g., Fractions, Geometry).

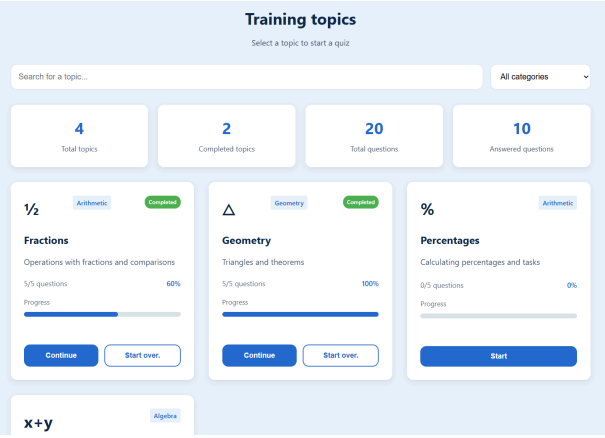


Fig. 2 Topic Selection

- **Learning Path Choice:** The student chooses between two modes:
  - **Interactive Lesson:** Engages with a dynamic visualization tool to explore concepts.
  - **Practice Quiz:** Takes an assessment on the selected topic.

Content Delivery & Interaction:

- **In Lesson mode:** the client loads the corresponding JSON file and renders the interactive components.

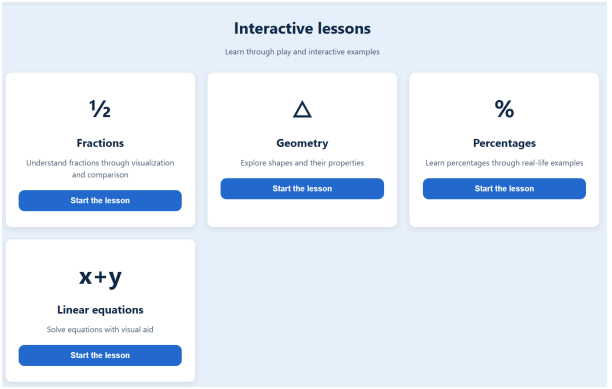


Fig. 3 Interactive Lessons Options

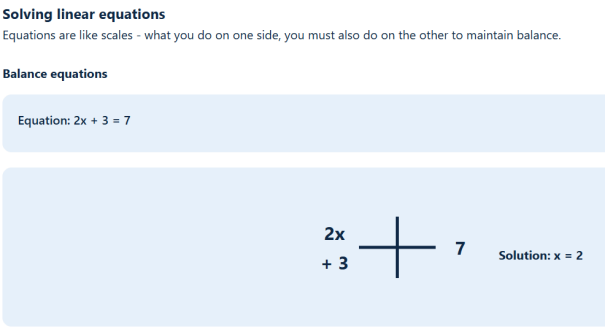


Fig. 4 Interactive Lesson Example – Linear equations

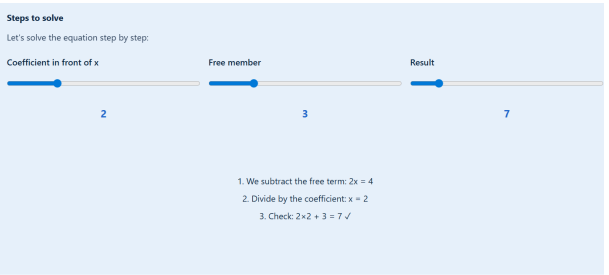


Fig. 5 Interactive Lesson Example – Linear equations

- **In Quiz mode:** questions are loaded from JSON, and answers are submitted and validated.

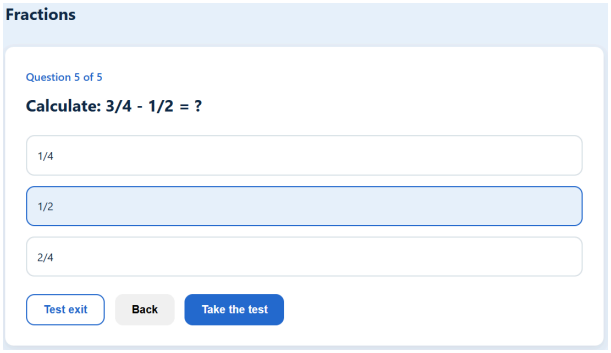


Fig. 6 Quiz Example

Data Persistence & Feedback: Quiz results, including specific mistakes, are written to the Firebase database. The client-side UI is immediately updated to reflect the new progress, providing instant feedback to the student.

Progress Visualization: The student can review their performance at any time on a dedicated summary page, which fetches and visualizes their historical data from Firebase.

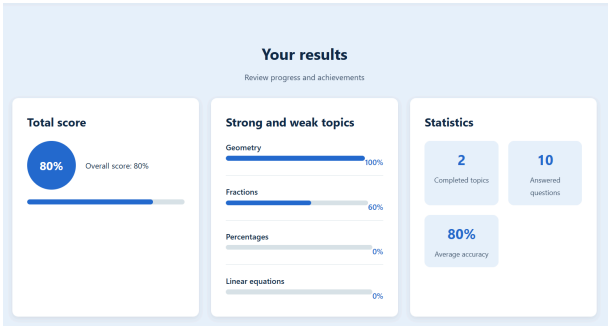


Fig. 7 Summary Page

4. IMPLEMENTATION

The implementation leverages modern web standards to create a responsive application experience.

4.1 TECHNOLOGY STACK

Frontend: HTML5, **CSS3** (with CSS Grid/Flexbox layout), and **JavaScript**. This choice ensures broad compatibility and fast loading times.

Backend & Services: Firebase Authentication (for user management), Firebase Realtime Database (for persistent, real-time data storage).

Hosting: The static files are hosted on **GitHub Pages**, demonstrating a zero-cost, high-availability deployment model.

4.2 CORE FUNCTIONAL COMPONENTS

4.2.1 AUTHENTICATION AND USER ROLE MANAGEMENT

User authentication is the gatekeeper for protected data. The following code fragment shows the user registration logic, which creates a user in Firebase Auth and initializes their data structure in the Realtime Database.

// Code Fragment: User Registration with Firebase

```
async signUp(email, password, name, role) {
    try {
        // 1. Create user with Firebase Authentication
        const userCredential = await auth.createUserWithEmailAndPassword(email, password);
        const user = userCredential.user;
        // 2. Update user profile
        await user.updateProfile({
            displayName: name
        });
        // 3. Save user data to database
        await database.ref('users/' + user.uid).set({
            role: role,
            name: name,
            email: email,
            createdAt: Date.now()
        });
        return { success: true, user: user };
    } catch (error) {
        return { success: false, error: error.message };
    }
}
```

Security is enforced through Firebase Realtime Database Rules, which prevent unauthorized access:

// Firebase Database Security Rules

```
{
 "rules": {
  "users": {
   "$uid": {
    // Users can only read/write their own data
    ".read": "$uid === auth.uid",
    ".write": "$uid === auth.uid"
   }
  },
  "topics": {
   ".read": true,
   ".write": false
  }
 }
}
```

4.2.2 JSON-BASED CONTENT STRUCTURE

Lessons and quizzes are defined in separate JSON files for modularity. This structure allows non-programmers to contribute content.

```
// Example: data/questions/fractions.json

{
 "topic": "fractions",
 "title": "Дроби",
 "description": "Операции с дроби и сравнения",
 "totalQuestions": 5,
 "questions": [
  {
   "id": 1,
   "question": "Кое е по-голямо: 3/5 или 4/7?",
   "options": ["3/5", "4/7", "Равни са"],
   "correctAnswer": 0,
   "explanation": "3/5 = 0.6, а 4/7 ≈ 0.57, следователно 3/5 е по-голямо."
  }
 ]
}
```

4.2.3 PROGRESS TRACKING AND DATA PERSISTENCE

A core feature is the detailed tracking of student performance. After a quiz is completed, the results are saved, including a log of specific mistakes for future remediation.

```
// Code Fragment: Saving Quiz Results to Firebase

async saveResults(score, mistakes) {
  // Retrieve existing user progress from localStorage, or start with an empty object
  const userProgress = JSON.parse(localStorage.getItem('userProgress') || '{}');

  // Update progress for the current topic
  userProgress[this.currentTopic] = {
    score: score,
    completedQuestions: this.questions.length,
    answers: this.userAnswers,
    mistakes: mistakes,
    completedAt: Date.now()
  };

  // Save updated progress back to localStorage
  localStorage.setItem('userProgress', JSON.stringify(userProgress));

  // If the user is logged in, save the results to Firebase
  if (typeof authManager !== 'undefined' && authManager.currentUser) {
    try {
      await database.ref(`users/${authManager.currentUser.uid}/scores/${this.currentTopic}`).set({
        score: score,          // Store score in Firebase
        mistakes: mistakes,
        completedAt: Date.now()
      });
    } catch (error) {
```

```
            // Log an error if saving to Firebase fails
            console.error('Грешка при запазване на резултатите:', error);
        }
    }
}
    // Code Fragment: Fetching and Visualizing Progress
async loadUserProgress() {
    // If this is a retake, reset progress to start over
    if (this.isRetake) {
        this.currentQuestionIndex = 0;          // Start from the first question
        this.userAnswers = [];                  // Clear previous answers
        return;
    }
    // Initialize variable to hold progress data
    let topicProgress = null;
    // Attempt to load progress from Firebase if the user is logged in
    if (typeof authManager !== 'undefined' && authManager.currentUser) {
        try {
            const snapshot = await database.ref(`users/${authManager.currentUser.uid}/scores/${this.currentTopic}`).once('value');
            topicProgress = snapshot.val();  // Retrieve saved progress for the current topic
        } catch (error) {
            // Log an error if Firebase retrieval fails
            console.error('Грешка при зареждане от Firebase:', error);
        }
    }
    // If no progress found in Firebase, fallback to localStorage
    if (!topicProgress) {
        const userProgress = JSON.parse(localStorage.getItem('userProgress') || '{}');
        topicProgress = userProgress[this.currentTopic];  // Retrieve topic progress from localStorage
    }
    // If user has previous progress, resume from the last completed question
    if (topicProgress && topicProgress.completedQuestions > 0) {
        // Ensure the index does not exceed total questions
        this.currentQuestionIndex = Math.min(topicProgress.completedQuestions, this.questions.length - 1);
 // Load saved answers
        this.userAnswers = topicProgress.answers || [];
    }
}
```

5. EXPERIMENTAL RESULTS

While a controlled study is required for definitive quantitative results, the implementation and initial use of SolveWise yield several observable outcomes and qualitative insights.

5.1 TECHNICAL PERFORMANCE

The platform successfully delivers on its core technical objectives. The client-server architecture ensures that user progress is consistently saved. The use of static JSON for content allows for rapid loading and easy maintenance. The Firebase integration provides authentication and real-time

data capabilities without the need for a custom backend server.

5.2 LIMITATIONS AND OBSERVATIONS

The current implementation has limitations. The adaptation is student-directed rather than algorithmically driven. The platform suggests areas of weakness but does not automatically serve new content based on mistake patterns. Furthermore, the content is currently limited to four core topics, and the platform lacks collaborative features or advanced teacher dashboards for classroom management.

6. COMMENTS AND CONCLUSIONS

SolveWise demonstrates that a significant portion of the benefits of adaptive learning technology can be achieved with a relatively simple, well-architected system. Its success hinges not on complex AI but on thoughtful pedagogical design - using interactivity to build intuition and transparency to encourage metacognition.

The primary challenge encountered was designing data structures that are both efficient for the database and rich enough for meaningful learning analytics. The decision to store entire quiz attempts, including mistake objects, was crucial for enabling future enhancements like personalized review sessions.

SolveWise provides a functional, effective, and scalable model for a digital mathematics remediation platform for middle school students. It successfully leverages modern web technologies to create an engaging learning environment that emphasizes conceptual understanding through interaction and self-paced progress. The platform stands as evidence that powerful educational tools can be built with lightweight, accessible technologies.

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